

Phase 4 Business Report

Executive Summary

The saved model is a Logistic Regression classifier. On the held-out test set it achieved $F1 = 0.6752$, $Recall = 0.8172$, and $ROC-AUC = 0.8129$. That makes it a strong decision-support tool for targeting retention spend.

The model ranking shows that Logistic Regression is the best on held-out test F1 in this run, while the tuned tree model and stacking ensemble trail behind on F1 and recall.

Customer risk is concentrated into three tiers: High Risk, Medium Risk, and Low Risk. The business should focus immediate action on the High Risk tier and use lighter-touch nudges for the Medium Risk tier.

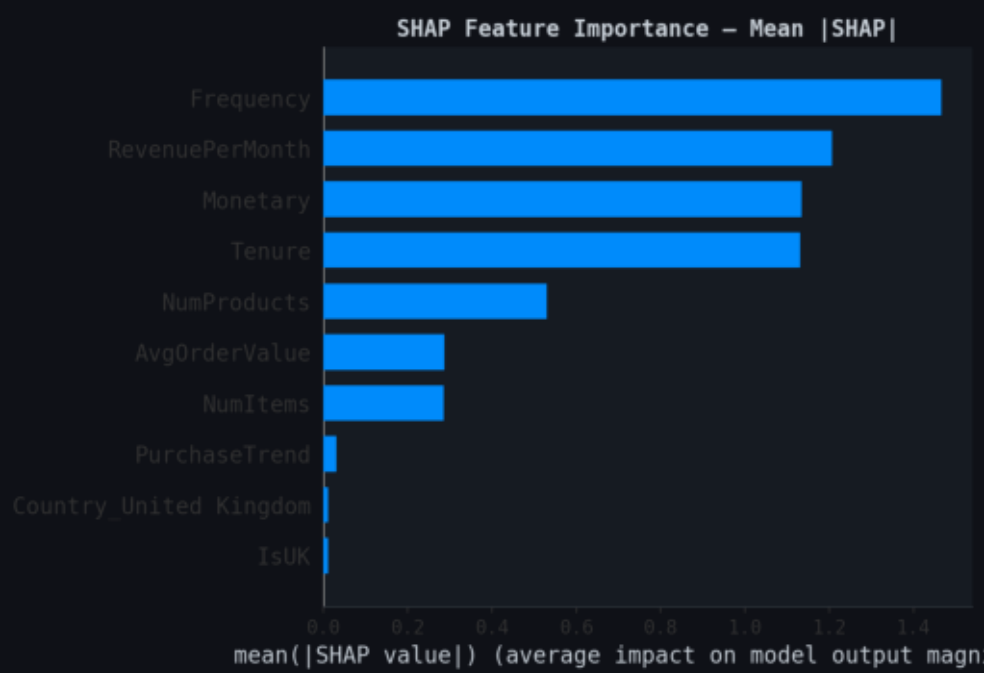
Metric	Value
F1	0.6752
Recall	0.8172
ROC-AUC	0.8129
High Risk customers	196
Medium Risk customers	289
Low Risk customers	383

Visual summary



Key Churn Drivers

SHAP highlights the features that move an individual customer toward or away from churn.



Top 5 drivers:

- Frequency: 1.4668 mean |SHAP|
- RevenuePerMonth: 1.2059 mean |SHAP|
- Monetary: 1.1342 mean |SHAP|
- Tenure: 1.1304 mean |SHAP|
- NumProducts: 0.5294 mean |SHAP|

Plain-English readout: customers who buy more often, spend more, and stay longer are less likely to churn; a weakening purchase pattern points in the opposite direction.

Customer Segments

Risk tiers translate the score into a simple campaign list.

RiskTier	Recency	Frequency	Monetary	AvgOrderValue
Low Risk	39.28	181.17	4365.10	27.15
Medium Risk	113.80	41.61	1052.97	304.57
High Risk	163.52	12.32	257.30	34.38



Segment sizes:
High Risk: 196 customers
Medium Risk: 289 customers
Low Risk: 383 customers

Recommendations, Ethics, and Roadmap

H. Call or email High Risk customers within 24 hours.

Why: This tier has the highest predicted churn probability and the strongest warning signals.

Estimated £ impact: Up to £45427

H. Automate a lighter nurture journey for Medium Risk customers.

Why: The model shows meaningful churn risk, but not enough to justify a full intervention.

Estimated £ impact: Up to £49982

M. Strengthen onboarding and early-life engagement for new or short-tenure customers.

Why: Customers with short histories are harder to score and can look artificially risky.

Estimated £ impact: Up to £62023

M. Calibrate thresholds or build a light regional check for non-UK customers.

Why: The portfolio is heavily UK-skewed, so smaller geographies may be less reliable.

Estimated £ impact: Up to £9097

Ethical considerations:

- The data is heavily UK-weighted, so small-country performance needs monitoring.
- New customers may be scored with less confidence because the model sees shorter histories.
- Mitigations: monitor country-level calibration and add cohort-aware review for early-life customers.

Implementation roadmap:

1. Week 1-2: launch High Risk outreach and tracking.
2. Month 1: automate Medium Risk nurture and monitor conversion.
3. Month 2: calibrate by country and add onboarding rules for new customers.
4. Ongoing: refresh SHAP and segment reports monthly.